Errata

Statistics for Chemical Engineers: From Data to Models to Decisions

Victor M. Zavala

1st edition (Cambridge University Press, 2025)

Last updated: November 3, 2026

This list outlines errata in the 1st edition of the book as printed by Cambridge University Press (2025); section, example, and equation numbers refer to the printed book. To report an error, please use the form on the book website.

Book

Location	**Error**	**Correction**
Eq. (1.5)	$\frac{\partial \log K_{eq}}{\partial T} = -\frac{\Delta H_0}{R \cdot T}$	$\frac{\partial \log K_{eq}}{\partial T} = \frac{\Delta H_0}{R \cdot T^2}$ (van 't Hoff equation)
Sections 1.2, 1.3, 2.1	Poisson domain $\{0, 1, 2, \ldots, \infty\}$	$\{0, 1, 2, \ldots\}$
Example 2.2	"…the machine has not failed in 79 out of 156 weeks."	"…the machine has failed in 79 out of 156 weeks." (consistent with $\hat{\mathbb{P}}(X = 0) = 79/156$, where $X = 0$ denotes failure)
Example 2.4	"…for a Bernoulli model, we must have that $\mathbb{P}(X = 0) = p$ and $\mathbb{P}(X = 1) = 1 - p$."	"…we must have that $\mathbb{P}(X = 0) = 1 - p$ and $\mathbb{P}(X = 1) = p$."
Eq. (2.14)	$\mathbb{V}_X = n \cdot p \cdot (1 - p) \cdot \frac{N_t - n}{n - 1}$	$\mathbb{V}_X = n \cdot p \cdot (1 - p) \cdot \frac{N_t - n}{N_t - 1}$
Example 2.6	$\mathbb{P}(X = 5) = f(5) = 0.1312$	$\mathbb{P}(X = 5) = f(5) = 0.0312$
Example 2.6	$X \sim \mathcal{H}(N_y, N_t, p, n)$	$X \sim \mathcal{H}(N_y, N_t, n, p)$ (same parameter order as in the definition of the hypergeometric RV)
Discussion of Figure 2.5	"the pdf of the RV $\mathcal{H}(\mathbf{y}, N, n, p)$ for $N = 1000$, …"	"the pdf of the RV $\mathcal{H}(N_y, N_t, n, p)$ for $N_t = 1000$, …"

Location	Error	Correction
Section 2.2	"…will shrink/damp uncertainty if $b < 1$ and will magnify uncertainty if $b > 1$."	"…if $\|b\| < 1$ …if $\|b\| > 1$."
Exercise 2.6	"…estimated to be 15 ± 1 MWh. The daily energy demand is assumed to be Gaussian and the confidence intervals correspond to a probability level of 95%." The meaning of ± 1 is ambiguous (it can be read as a standard deviation of 1 MWh).	"…estimated to be 15 ± 1 MWh, where ± 1 MWh is the half-width of the 95% confidence interval (i.e., the 95% confidence interval of the daily energy demand is $[14, 16]$ MWh). The daily energy demand is assumed to be Gaussian."
Example 3.2	$(1 - 0.36) \cdot (1 - 0.90) = 0.064$	$(1 - 0.37) \cdot (1 - 0.90) = 0.060$
Eq. (3.29)	Last term: $+\mathbb{P}(X_1 \leq x_1 + \Delta x_1 \ \& \ X_2 \leq x_2)$	$-\mathbb{P}(X_1 \leq x_1 + \Delta x_1 \ \& \ X_2 \leq x_2)$
Example 3.16	$\Sigma_{2\|1} = 0.015$; "relative reduction of about 25%" (and of 21% for $\Sigma_{1\|2}$)	$\Sigma_{2\|1} = 0.016$; relative reduction of about 22% in both cases
Eq. (3.114)	$f(x_1, x_2) = \dfrac{1}{2\pi\sqrt{\Sigma_{11}\Sigma_{22}}} \exp(-E(x_1, x_2))$	$f(x_1, x_2) = \dfrac{1}{2\pi\sqrt{(1-\rho^2)\Sigma_{11}\Sigma_{22}}} \exp(-E(x_1, x_2))$
Example 4.1	"From the definition of the first central moment, we have $\mathbb{CMO}_1(\theta) = \mathbb{E}[(X - \mathbb{E}_X)] = \mathbb{E}_X - \mu = 0$."	"From the definition of the first raw moment, we have $\mathbb{RMO}_1(\theta) = \mathbb{E}_X = \mu$."
Example 4.2	$\hat{\mathbb{V}}_X = \beta^2 \left(\Gamma(1 + 2/\xi) + \Gamma(1 + 1/\xi)^2\right)$	$\hat{\mathbb{V}}_X = \beta^2 \left(\Gamma(1 + 2/\xi) - \Gamma(1 + 1/\xi)^2\right)$
Examples 4.6 and 4.7	$X \sim$ Weibull$(2, 1)$	$X \sim$ Weibull$(1, 2)$, that is, shape $\xi = 1$ and scale $\beta = 2$ in the notation Weibull(ξ, β) (this is consistent with $\mathbb{E}[X] = 2$ and $\mathbb{V}[X] = 4$ reported in Example 4.7)
Section 4.3	"…can achieved by using a powerful result…"	"…can be achieved by using a powerful result…"
Section 5.1	"…they will sit at a shaper minimum…"	"…they will sit at a sharper minimum…"
Section 5.3	"…θ are the kinetic constraints."	"…θ are the kinetic constants."

Location	Error	Correction
Example 5.10	"…lands at a minimum point that has nonnegative physical values."	"…lands at a minimum point that has negative (nonphysical) values."
Below Eq. (5.115)	"This penalizes squared deviations from a reference value $\hat{\theta}$."	"This penalizes squared deviations from a reference value $\bar{\theta}$."
Section 5.4	"(we will see this is Section 5.5)"	"(we will see this in Section 5.5)"
Example 5.12	$SSE(\theta) + \frac{1}{2}\|\|\theta - \bar{\theta}\|\|_2^2$	$SSE(\theta) + \kappa \cdot \frac{1}{2}\|\|\theta - \bar{\theta}\|\|_2^2$ (with $\kappa = 1$, as stated in the example)
Example 5.13	$\hat{\theta} = (0.25, 0.06, 0.04, 0.11, 1.19)$	$\hat{\theta} = (1.25, 0.06, 0.04, 0.11, 1.19)$ (this coincides with the estimates obtained in Example 5.10)
Eq. (6.4)	$\hat{\text{Cov}}(X) = \mathbf{X}^T\mathbf{X}$	$\hat{\text{Cov}}(X) = \frac{1}{S}\mathbf{X}^T\mathbf{X}$
Eq. (6.42)	$g_j = e^{-j^2/\sigma^2}$	$g_j = e^{-j^2/(2\sigma^2)}$ (the weights reported in Examples 6.9 and 6.10 use this form)
Example 6.10	$g = (0.0003, 0.41, 0.41, 0.003)$	$g = (0.0003, 0.41, 0.41, 0.0003)$
Section 6.3, inverse FT	$f(t) = \int_{-\infty}^{\infty} \hat{f}(\omega) e^{i(2\pi\omega\tau)} d\omega$	$f(t) = \int_{-\infty}^{\infty} \hat{f}(\omega) e^{i(2\pi\omega t)} d\omega$
Eq. (6.76)	$g(x,y) = \frac{1}{\sqrt{2\pi\sigma^2}} e^{-\frac{x^2+y^2}{2\sigma^2}}$	$g(x,y) = \frac{1}{2\pi\sigma^2} e^{-\frac{x^2+y^2}{2\sigma^2}}$
Section 6.5	"…a Gaussian prior $\theta \sim \mathcal{N}(0, \lambda\mathbf{I})$"	"…a Gaussian prior $\theta \sim \mathcal{N}(0, \lambda^{-1}\mathbf{I})$"
Eq. (6.123)	$\ldots - \frac{1}{2}\mathbf{y}^T\mathbf{C}(\gamma)\mathbf{y}\ldots$	$\ldots - \frac{1}{2}\mathbf{y}^T\mathbf{C}(\gamma)^{-1}\mathbf{y}\ldots$
Eq. (6.124)	$\sigma^2(x) = c(x,\hat{\gamma}) - \mathbf{k}(x,\hat{\gamma})^T\mathbf{C}(\hat{\gamma})\mathbf{k}(x,\hat{\gamma})$	$\sigma^2(x) = c(x,\hat{\gamma}) - \mathbf{k}(x,\hat{\gamma})^T\mathbf{C}(\hat{\gamma})^{-1}\mathbf{k}(x,\hat{\gamma})$
Example 7.5	"Standard deviation: $\mathbb{SD}[Y(u)] = 19{,}284$ USD, $\mathbb{V}[Y(u')] = 79{,}288$ USD."	"Standard deviation: $\mathbb{SD}[Y(u)] = 19{,}284$ USD, $\mathbb{SD}[Y(u')] = 79{,}288$ USD."
Example 7.5	"…a risk-conscious DM would prefer u because it has a lower worst-case value…"	"…a risk-averse DM would prefer u because it has a lower worst-case value…"
Eq. (7.33)	$\min_{u\in\mathcal{U}} \mathbb{P}(\varphi(X,u) > y)$	$\min_{u\in\mathcal{U}} \mathbb{P}(\varphi(X,u) > \bar{y})$

Location	Error	Correction
Section 7.7	"…we augment our dataset $\mathcal{D}_{\ell+1} \leftarrow \mathcal{D}_\ell \cup (u_\omega, y_\omega)$."	"…we augment our dataset $\mathcal{D}_{\ell+1} \leftarrow \mathcal{D}_\ell \cup \{(u'_\omega, y'_\omega)\}$."

Code, data, and slides

Location	Error	Correction
Exercise 1.7, `clinical_largegroup.dat`	The data file for the large groups (100 people) is identical to `clinical_smallgroup.dat`; the script `ch1_clinical_trials.m` saved the small-group data to both files.	Corrected: the script now saves the large-group data to `clinical_largegroup.dat`, and the data file has been regenerated (100 people, five trials).
Example 3.20 (e-book link)	The link to `ch3_gaussian_samples_ellipse.m` points to a misspelled file name (`ch3_gaussian_samples_eliipse.m`).	Link corrected.
Example 6.3, `ch6_gibbs_pca_estimation_sparse.ipynb`	The Python notebook does not reproduce the eigenvector matrices W reported in the example.	The notebook now uses the same sparse PCA method as the book and reproduces the informative columns of W and $\tilde{W}$. The input matrix has rank 3, so the last three eigenvalues are zero and the corresponding columns of W are not unique (they can differ across numerical libraries).
Example 6.26, `ch6_gibbs_neuralnet.ipynb`	The Python notebook produces results that differ substantially from those reported in the example.	Fixed in the October 2026 version of the notebook: the NN now fits the data as well as the GP model (mean squared error of about 9×10^{-4} for both), as reported in the example.
Chapter 3 slides	Discrete domain $\mathcal{D} = \{(l_1, l_2), (l_1, u_2), (u_1, l_2), (u_2, l_2)\}$.	$\mathcal{D} = \{(l_1, l_2), (l_1, u_2), (u_1, l_2), (u_1, u_2)\}$ (this is correct in the printed book).

Acknowledgments

I thank the following readers for reporting errors: Ethan Deutsch, Jadiel Lopez Gonzalez, and Joshua Pulsipher.